

ABDERAHMAN EL FARCHOUNI

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SUMMARY

PhD candidate in Hydrology, isotopes, hydrochemistry and geophysics at University Mohammed VI Polytechnique (UM6P - IWRI) and Università degli Studi di Firenze (UNIFI - DAGRI) through a cotutelle program. My research focuses on groundwater recharge and water resources management, combining hydrological modeling, remote sensing, geophysics, hydrochemistry, isotopes, and machine learning to improve hydrogeological conceptual models in semi-arid regions. I am experienced in both fieldwork and data analysis, with the ability to design multidisciplinary research frameworks that support sustainable water governance.

EDUCATION

PhD

Oct 2023 - Present

- Research focus: Underground hydraulic aquifer systems of the piedmont of the Atlas of Marrakech : hydrogeophysical assessment and modeling

Master

2021-2023

Faculty of Sciences, Kenitra

- Master Hydroinformatic and Hydrosystems Management

Bachelor in Applied Geosciences

2017-2021

Faculty of Science and Technology, Tangier – Morocco

- Applied Geosciences

DEUST

2017-2020

Faculty of Science and Technology, Tangier – Morocco

- Foundation degree (Bac+2) providing multidisciplinary scientific background.

Baccalaureate

2020

- Baccalaureate in Mathematics A

Baccalaureate

2017

- Life and Earth Sciences

RESEARCH EXPERIENCE

PhD Researcher – Hydrology, hydrochemistry, isotopes and Hydrogeophysics

Oct 2023 - Present

Mohammed VI Polytechnic University (UM6P), International Water Research Institute (IWRI)

- Conducting research on groundwater recharge and conceptual modeling of the Haouz–Mejjate aquifer system (Atlas of Marrakech).
- Integration of hydrochemistry, stable isotopes, geophysics, and remote sensing to assess recharge dynamics in semi-arid regions.
- Multidisciplinary framework combining field campaigns, data analysis, and hydrological modeling.
- Integrated Hydrological Modelling

RESEARCH EXPERIENCE

Fieldwork and Data Collection:

- **Geophysical surveys:** Electrical Resistivity Tomography (ERT), Magnetic Resonance Sounding (MRS), seismic surveys, and EM38.
 - **Hydrogeological monitoring:** Piezometric network measurement and data analysis.
 - **Water sampling:** Collection and analysis of isotopic ($\delta^2\text{H}$, $\delta^{18}\text{O}$) and hydrochemical samples.
 - **Aquifer characterization:** Execution and interpretation of pumping tests.
 - **Socio-economic surveys:** Design and implementation of farmer questionnaires on irrigation practices, groundwater recharge, and water management strategies.
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PUBLICATIONS

Peer-Reviewed Journal Articles:

[1] El Farchouni, A., Hadri, A., Fakir, Y., Ouarani, M., Azaroual, M., & Kchikach, A.. "Mapping groundwater recharge potential zones in a semi-arid, anthropogenically modified mountainous basin." *Scientific African*, 30(October), e03025. <https://doi.org/10.1016/j.sciaf.2025.e03025>, 2025.

[2] El Mezouary, L., Elfarchouni, A., Hadri, A., Hakim, M., Fakir, Y., Bouchaou, L., & Chehbouni, A.. "Geostatistical modeling of subsurface heterogeneity in the Al-Haouz-Mejjate aquifer system, Morocco: A T-PROGS approach for enhanced hydrogeological characterization." *Scientific African*, 31, e03163. <https://doi.org/10.1016/j.sciaf.2025.e03163>, 2026.

[3] El Mezouary, L., Kharrou, M. H., Hadri, A., Elfarchouni, A., Fakir, Y., Bouchaou, L., & Chehbouni, A.. "Advanced hydrogeological modeling to reduce uncertainty in groundwater recharge estimation in semi-arid aquifers: Case of Al-Haouz-Mejjate aquifer in Morocco." *Chemosphere*, 392, 144743, 2025.

[4] El Mezouary, L., Hadri, A., Kharrou, M. H., Fakir, Y., Elfarchouni, A., Bouchaou, L., & Chehbouni, A.. "Contribution to advancing aquifer geometric mapping using machine learning and deep learning techniques: a case study of the AL Haouz-Mejjate aquifer, Marrakech, Morocco." *Applied Water Science*, 14(5).

Conference Proceedings

[5] El Mezouary, L., Hadri, A., Kharrou, M.H., Fakir, Y., Elfarchouni, A., Bouchaou, L., Chehbouni, A.. "Machine Learning and Deep Learning Guided Assessment of Groundwater Reservoir Hydrodynamic Parameters: A Case Study of The El Haouz Aquifer." 4th International GIRE3D Congress on Participatory and Integrated Management of Water Resources in Arid Zones, Laayoune, Morocco, Nov 2023.

[6] El Farchouni, A., et al.. "Mapping Groundwater Recharge Potential Zones Using Remote Sensing, GIS, and AHP Techniques: A Case Study of the Tensift Basin, Morocco." *IAHR Africa Congress*, Dec 9–12, 2024, 2024.

TECHNICAL SKILLS

Remote Sensing & GIS:

- **Google Earth Engine (GEE)** for large-scale geospatial analysis (land use/land cover, NDVI/NDWI, evapotranspiration modeling, groundwater recharge mapping).
- **ArcGIS, QGIS** for spatial analysis, interpolation, and cartography.
- Satellite data processing (Sentinel-2, Landsat, CHIRPS, WaPOR, TerraClimate, SMAP, GPM etc).

Hydrological & Hydrogeological Modeling:

- **SWAT/SWAT+ and MODFLOW** for integrated surface-groundwater modeling
- **GMS:** Groundwater Flow modeling

Geophysics & Field Methods:

- Electrical Resistivity Tomography (ERT), Magnetic Resonance Sounding (MRS), seismic surveys, EM38.
- Pumping tests, piezometric monitoring, isotopes ($\delta^2\text{H}$, $\delta^{18}\text{O}$) and hydrochemistry sampling.
- Geophysics software: Prodiviner, Res2DInv

Programming & Data Science:

- **Python** (pandas, scikit-learn, TensorFlow, Xarray, matplotlib etc).
- **Machine learning & deep learning** (KNN, Random Forest, LSTM, GAIN, PINN)
- **R for statistical and hydrogeochemical analysis** (hydrogeochemistry diagrams, PCA, clustering).
- **SQL & SQLite Studio** for database management.

Other Tools:

- **Adobe Illustrator** for scientific figure editing and publication-quality graphics.
- **Microsoft Office** Suite (Excel, Word, PowerPoint).
- **LaTeX** for academic writing and reports.

CONFERENCES WORKSHOPS & TRAINING

- **IAHR Africa Congress:**
 - Mapping Groundwater Recharge Potential Zones Using Remote Sensing, GIS, and AHP Techniques: A Case Study of the Tensift Basin, Morocco (Dec 9–12, 2024).
- **Discover Scopus AI (May 15, 2024).**
- **Workshop on Isotopic Tracers in Hydrology and Earth Sciences** (May 28–29, 2024).
- **Karst and Cave Sustainable Management Summer Course** (Jul 29 – Aug 9, 2024).
- **Enhancing Water Productivity with WaPOR: Using WaPLUGIN in QGIS** (Dec 12, 2024).
- **Mathematical Models and Artificial Intelligence for Solving Groundwater Flow Problems** (Jan 6–7, 2025).
- **Maghrebien Colloquium of Applied Geophysics (MCGA-9) – Active participation in the 9th edition, Faculty of Sciences, Ibn Tofail University, Kenitra, Morocco** (Apr 23–25, 2024).
- **SYMPLE: Introduction to Applied Groundwater Flow Modelling with GW Vistas – online training** (Mar 17–19 & Apr 7–10, 2025).
- **SYMPLE: Understand MODFLOW** (May 6–27, 2025).

LANGUAGES

- **Arabic (Native)**
- **French (Fluent)**
- **English (Fluent)**